

1. Consider a scalar field theory described by the interaction lagrangian $L = -g\phi^3 - \lambda\phi^4$, where λ and g are coupling constants. What are the Feynman rules for the vertices of the theory? Write down all the topologically inequivalent diagrams that contribute to the process $\phi(\vec{p}_1) + \phi(\vec{p}_2) \rightarrow \phi(\vec{p}_3) + \phi(\vec{p}_4)$ to lowest non-trivial order in the couplings g and λ . Write down the corresponding S matrix element.
2. The scattering amplitudes should be unchanged if we make a field redefinition. Suppose, for example, we have, $L = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2$ and we make the field redefinition $\phi \rightarrow \phi + \lambda\phi^2$. Work out the lagrangian in terms of the redefined field, and the corresponding Feynman rules. Compute (at tree level) the $\phi\phi \rightarrow \phi\phi$ scattering amplitude. You should get zero, because this is a free-field theory in disguise.